Volatility Regime Analysis of Brent Crude Oil Prices Using Discrete-Time Markov Chains

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Abstract

This paper applies a Markov chain framework to classify and analyze volatility regimes in daily Brent crude oil prices from February 1990 to April 2025. Three regimes (low, medium and high) are defined using thresholds on standard deviation of daily returns. Results show that all three regimes are highly self-persistent, with self-transition probabilities ranging from 0.910 (Medium) to 0.962 (High). Further analysis shows expected regime durations of 11 to 27 trading days. Historical data is augmented with live data from the U.S. Energy Information Administration (EIA) to identify the current regime and forecast its expected remaining duration.

1 Introduction

Crude oil is among the most economically significant commodities in the world. Fluctuations in its price propagate through the market at large and have been the focal point of past and present geopolitical action. Understanding the statistical structure of these fluctuations, especially the tendency for high and low volatility periods to cluster in time, has direct implications for risk management, policy analysis, and investment strategy.

Regime models have a long tradition in financial econometrics. Hamilton (1989) introduced the hidden Markov model for business cycle analysis, demonstrating that macroeconomic

time series often exhibit distinct, persistent states separated by abrupt transitions [1]. Subsequent work extended this framework to commodity markets, including crude oil [2, 3].

The remainder of this paper is organized as follows. Section 2 describes the data and methodology. Section 3 presents the estimated transition matrix, regime persistence statistics, and live regime forecast. Section 4 discusses the findings and their implications. Section 5 concludes the paper with summarized findings and potential follow-on work related to the analysis presented here.

2 Methods

2.1 Data

Daily Brent crude oil spot prices were sourced from a compiled dataset spanning January 1990 to April 2025. For live regime classification, contemporaneous Brent spot prices are retrieved via the EIA Open Data API [4]. The key variable used for regime assignment is `brent_30d_vol`, defined as the 30-trading-day rolling standard deviation of the daily percentage return:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}} \times 100, \qquad \sigma_t = \sqrt{\frac{1}{29}\sum_{i=0}^{29}(r_{t-i} - \bar{r})^2}$$

where P_t is the Brent price on day t. The first 40 observations (where $\sigma_t = 0$) and a constant values appended after April 3, 2025 are excluded, leaving $n = 7{,}419$ trading days.

2.2 Regime Definition

Three volatility regimes are defined using the 33rd and 67th percentiles of σ_t over the full sample, yielding roughly equal-sized states:

$$\text{regime}_t = \begin{cases} \textsc{Low} & \text{if } \sigma_t < 1.51 \\ \textsc{Medium} & \text{if } 1.51 \leq \sigma_t < 1.89 \\ \textsc{High} & \text{if } \sigma_t \geq 1.89 \end{cases}$$

2.3 Markov Chain Estimation

Let $S = \{\textsc{Low}, \textsc{Med}, \textsc{High}\}$ describe the state space. The one-step transition counts n_{ij} are obtained by tallying consecutive-day regime pairs over the full dataset. The maximum-

likelihood estimate of each transition probability is:

$$\hat{p}_{ij} = \frac{n_{ij}}{\sum_k n_{ik}}$$

Row normalization ensures $\sum_j \hat{p}_{ij} = 1$ for all i. The resulting matrix $\hat{\mathbf{P}}$ is the primary focus for analysis.

2.4 Stationary Distribution and Regime Duration

The expected duration of a single stint in regime i follows a geometric distribution with success probability $(1 - \hat{p}_{ii})$, giving:

$$\mathbb{E}[\text{duration}_i] = \frac{1}{1 - \hat{p}_{ii}}$$

By the Markov memoryless property, the expected *remaining* time in the current regime equals this quantity regardless of how many days the current stint has already lasted.

3 Results

3.1 Transition Probability Matrix

Table 1 presents the estimated 3×3 transition matrix. All diagonal entries exceed 0.91, confirming strong persistence within that regime. Off-diagonal entries show a sequential structure where transitions between Low and High almost never occur directly ($\hat{p}_{\text{Low,High}} = 0.0016$, $\hat{p}_{\text{High,Low}} = 0.0004$); the market instead passes through Medium volatility as an intermediate state.

Table 1: Estimated one-step transition probability matrix.

	Low	Medium	High
Low	0.9428	0.0556	0.0016
Medium	0.0547	0.9104	0.0349
High	0.0004	0.0372	0.9624

3.2 Regime Persistence

Table 2 summarizes the distribution of consecutive-day run lengths for each regime. The High volatility regime produces the longest average stints (26.6 days) despite having the

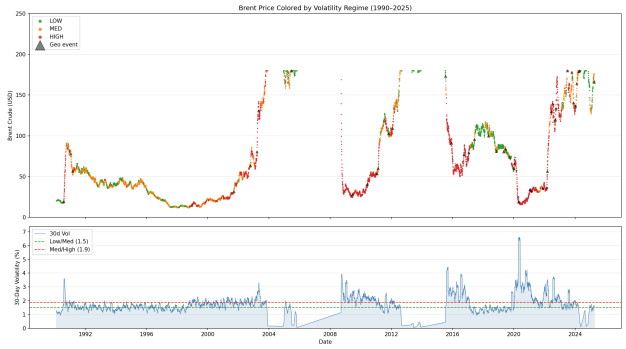


Figure 1: Regime Timeline

fewest distinct episodes (92). Medium volatility stints are the most numerous (227) but shortest on average (11.1 days).

Table 2: Regime persistence statistics (trading days per consecutive stint).

Regime	**Stints**	**Mean**	**Median**	**Std**	**Max**
Low	140	17.5	5	28.5	167
Medium	227	11.1	6	12.1	67
High	92	26.6	7	53.7	325

3.3 Stationary Distribution

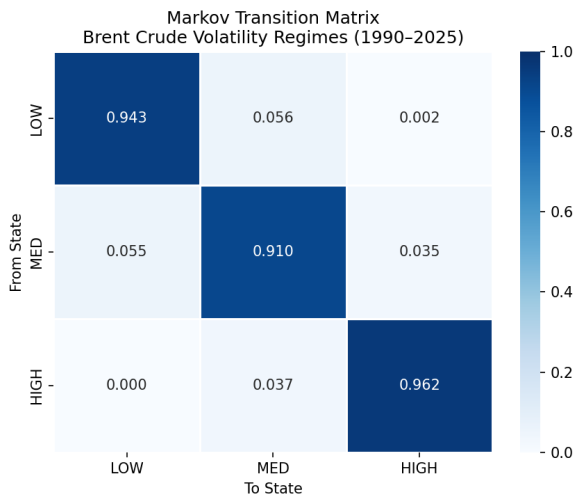
The stationary distribution is approximately uniform: $\pi_{\text{Low}} = 32.9\%$, $\pi_{\text{Med}} = 34.1\%$, $\pi_{\text{High}} = 33.1\%$. This shows no single volatility state dominates the long-run behavior of Brent crude.

3.4 Visualizations

The figure below displays the transition probability matrix as a heatmap. Figure 1 shows the full 35-year price history colored by volatility regime, with the 30-day rolling volatility and regime thresholds plotted below. There are some easily identifiable high-volatility episodes to include the 1990 Gulf War, the 1998 Asian financial crisis, the 2008 financial crisis, and the 2020 COVID-19 demand shock.

3.5 Live Regime Forecast

Using EIA spot price data retrieved on April 17, 2026, the current 30-day rolling volatility places Brent crude in the High volatility regime. The self-transition probability for High is



$\hat{p}_{\text{HIGH,HIGH}} = 0.9624$, implying an expected remaining duration of:

$$\mathbb{E}[\text{remaining}] = \frac{1}{1 - 0.9624} \approx 26.6 \text{ trading days}$$

4 Discussion

The estimated Markov chain captures several well-documented features of commodity volatility. First, consider volatility clustering where the tendency for large price movements to follow large movements manifests directly in the high diagonal entries of $\hat{\mathbf{P}}$. Second, the near-zero probability of direct low to high transitions supports a sequential escalation model of market stress. This is consistent with earlier volatility literature [1].

The large gap between mean (26.6 days) and median (7 days) for the High regime suggests positive duration dependence. Once a high-volatility episode extends beyond its median, it tends to persist significantly longer.

A limitation of this approach is the static threshold calibration. Because thresholds are fixed at sample quantiles, regime assignments do not adjust for secular trends in volatility levels.

5 Conclusion

This paper demonstrates that a simple, observable-state Markov chain provides a tractable and interpretable framework for modeling volatility regimes in Brent crude oil prices. The estimated transition matrix reveals strong within-regime persistence, near-zero direct low to high transitions, and an approximately uniform stationary distribution. High-volatility

regimes are the most persistent and have the widest duration distribution, with individual episodes lasting up to 325 trading days.

The integration of live EIA data enables real-time regime classification to extend the historical model into a practical monitoring tool. Future work could incorporate other data such as OPEC production or geopolitical event flags, or apply the framework to other crude benchmarks (WTI, Dubai) for cross-market comparison.

References

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